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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS**PURPOSE**

This document defines the analysis and design procedures for heat exchanger and horizontal vessel foundations.

SCOPE

This document addresses the loading conditions for which a heat exchanger or horizontal vessel foundation should be designed, and the design of each component. Two sample calculations are included for typical situations. The engineer should implement this method into the calculations for the project.

APPLICATION

This document may be used as the basis for design of any similarly supported type of equipment. Computer programs are available to automate this document. This should be discussed with the Lead Engineer at the start of the project.

GENERAL

Within this document, the term "vessel" is used to refer to either exchangers or horizontal vessels. Where criteria applies to one type of equipment or the other, more specific terminology is used.

The Structural Engineer should verify the following items from the exchanger drawings:

- Spacing of supports
- Width and length of the supports
- Vessel weights (refer to Vertical Loads below)
- Anchor bolt hole sizes and thickness of base plate
- Location of holes and slots in supports for lateral load analysis

Maximum design temperature or special conditions should be verified with the Lead Process Engineer. Typically, design temperatures provided by Process Department are 25 to 50 degrees F above normal operating conditions. Conditions such as steam-out, depressurizing, and regeneration may dictate even higher design temperatures.

The use of insulation pads between an exchanger and foundation should be considered for exchangers with operating temperatures below 32 degrees F and for exchangers with bottom of base plate temperatures greater than 500 degrees F.

Horizontal vessels and exchangers are usually supported by rectangular piers. If elevated above an economical height for construction of piers, other types of supports such as T-supports may be considered. Usually, the dividing line is approximately 12 feet above the top of the footing, or minimum headroom clearance under the beam. If T-supports are used, consideration should be given to the possibility of reinforcement congestion, especially in Seismic Zones 3 and 4 of the UBC (Uniform Building Code).

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The engineer should obtain all of the loading data, including the wind and earthquake loading, and then apply the applicable code formulas.

Vertical Loads

The following loads shall be considered as dead loads when applying load factors.

Erection Weight

This is the fabricated weight of the vessel, generally taken from the vessel drawing.

Empty Weight

This is the weight of the vessel (including exchanger tube bundles) generally taken from the vessel drawing, plus the weight of piping and insulation. Except for special cases, the weight of piping and insulation may be taken as 20 percent of the empty vessel's weight. For stacked exchangers, use 20 percent of one exchanger's weight.

If exchanger weights are unavailable from the mechanical engineer, Attachment 01 may be used to determine an estimated weight of exchanger and tube bundle.

Operating Weight

This is the empty weight plus the weight of operating fluid, plus the weight of piping and insulation. Except for special cases, the weight of piping and insulation may be taken as 20 percent of the operating vessel's weight. For stacked exchangers, use 20 percent of one exchanger's weight. Occasionally, it may be necessary to have operating conditions verified by the Process Department.

Test Weight

This is the empty weight plus weight of water required for hydrostatic test, generally taken from the vessel drawing.

Eccentric Load

Unless more exact information regarding piping supported on the foundation is available, a load of 10 percent of the applicable vessel weight should be applied to each pedestal and footing at a distance of $D/2$ plus 18 inches from the longitudinal centerline of the vessel, where "D" is the basic diameter (basic diameter = exchanger I.D. + 2 times the wall thickness + 2 times the insulation thickness).

Load Distribution (Exchangers)

For most common shell and tube heat exchangers, vertical loads should normally be distributed with 60 percent to the channel end (slotted end) support and 40 percent to the shell end (fixed end) support. However, the actual exchanger shape and support configuration should be reviewed when determining weight distribution, since in many

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cases load distribution may vary.

Horizontal Loads***Transverse Wind***

The wind pressure on the projected area of the side of the vessel should be applied as a horizontal shear at the center of the vessel. It is important not to forget the wind loading on projections such as piping, manways, insulation, and platforms during the wind analysis. For details of wind load calculations, refer to Structural Engineering Guideline 000.215.1215: Wind Load Calculation.

The saddle-to-pier connection should be considered fixed for transverse loads. The saddle design is a function of either the vessel supplier or the Mechanical Group.

Longitudinal Wind

The wind pressure on the end of the vessel should be applied as a horizontal shear at the center of the vessel. Also, include the flat surface wind pressure on the exposed area of both piers or columns, applied as a horizontal shear at the centroid of the exposed area.

The saddle-to-pier connection will be considered pinned for longitudinal loads.

Shielding

Make no allowance for shielding from wind by nearby equipment or structures except under unusual conditions.

Seismic

In earthquake zones, replace wind loads with seismic loads when calculations show that seismic loads are greater. For low friction slide plates ($\mu \leq 0.2$), apply all of the seismic force at the fixed pier. For higher friction slide plates ($\mu > 0.2$), apply 70% of the seismic force at the fixed pier. Distribute transverse and vertical seismic loads proportionately to both piers. Seismic pier forces shall be determined by using half the pier weight and by applying the resulting force at the top of the pier. Refer to Structural Engineering Guideline 000.215.1216: Earthquake Engineering, for details of seismic load calculations.

Bundle Pull (Exchangers)

Bundle pull force is an estimated friction force generated from exchanger tube bundle installation or removal. Unless job specifications specify otherwise, the bundle pull force should be taken as 1.5 times the tube bundle weight, but not less than the smaller of 2,000 pounds or the total weight of the exchanger. The load is applied longitudinally at the center of the bundle and distributed between the piers. For stacked exchangers, the load should be applied at the center of the top bundle. The portion of the bundle pull force at the sliding end support will equal the lessor of the friction force or 1/2 the bundle pull force. The remainder of the bundle pull force will be resisted at the fixed end support.

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Advise the Vessel Engineer of loads considered to be acting on the saddles.

Thermal Force

Thermal force is the force due to the growth of the horizontal vessel or exchanger between supports.

Calculate thermal growth using maximum design temperature. Thermal coefficients can be found in Attachment 02.

The thermal force used for design should be the smaller value resulting from the following two calculations:

- The force required to overcome static friction between the vessel support and the slide plate:

$$F_f = \mu (P_{op}) \quad \text{(Equation 1)}$$

where,

F_f = static friction force

μ = coefficient of friction, refer to the table under Slide Plates for values

P_{op} = operating compression load on slide plate

- The force required to deflect the pier or column an amount equal to 1/2 of the thermal growth between supports:

$$F_d = \frac{3 \Delta E I}{2 L^3} \quad \text{(Equation 2)}$$

where,

F_d = pier deflection force

Δ = total deflection between piers

E = modulus of elasticity

I = pier moment of inertia

L = distance between piers

The thermal force should be applied at the top of the piers.

Load Distribution

Divide the load equally between piers except where otherwise specified.

When the longitudinal loads, such as seismic and bundle pull, govern the pier/foundation design, a strut between the piers has been found to be economical. The savings in concrete commonly exceeds the added cost of a strut. A strut could consist of two nested pipes field welded to each other and to embedded plates on the inside of each pier. If the vessel or exchanger has a central pipe which interferes with a strut on the centerline, two struts can be used, which straddle the pipe.

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Load Combinations

Footings and piers should be designed to resist the load combinations listed in 000.215.00910, Structural Design Criteria.

Foundations for fin exchangers (double pipe exchangers) should not be designed to resist thermal or bundle pull forces.

Piping thermal loads should be included in combinations when deemed advisable and should be considered as dead loads when applying load factors.

A combination of erection weight plus wind should be used when deemed applicable by job criteria.

Design Stresses

An increase in allowable soil bearing pressure for short term loads should be based on job criteria.

ANCHOR BOLTS

Use headed bolts. Check anchor bolts in accordance with Structural Engineering Guideline 000.215.1207: Anchor Bolt Design Criteria. Friction force at the bottom of the saddle must be overcome before lateral load is assumed to produce shear in the anchor bolts. For seismic loads, apply horizontal shear forces to the anchor bolts.

SLIDE PLATES

As a minimum, provide a steel slide plate at the sliding end of every vessel regardless of the flexibility inherent in the structural support.

Use low friction slide plates such as lubrite or teflon for large movements or if the operating weight at the sliding end exceeds 25 kips. Exceptions to this should be as follows:

- When the deflection force F_d is equal to or less than the sliding force F_f for low friction bearing.
- When it is more economical to distribute a large load more evenly between the two piers by increasing the friction force on the sliding end.

Typical coefficients of friction are as follows:

- | | |
|--|------|
| • No slide plate (steel support on concrete) | 0.50 |
| • Steel Slide Plate | 0.30 |
| • Teflon Slide Plate | |
| – Bearing Pressure 100 psi and below | 0.10 |
| – Bearing Pressure Above 100 psi | 0.06 |
| • Lubrite Slide Plate | 0.10 |

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Use the following criteria for sizing low temperature (less than 500 degrees F) slide plate elements. For higher temperatures, refer to manufacturer's literature.

Plate width: (Δ = total thermal growth between piers)

- lower element = (saddle width) + 2 (Δ) + 1/4 in
- upper element = (lower element) + 1 in

Plate length: (use 18 inch maximum distance between plates)

- lower element = based on allowable contact pressure & width
- upper element = (lower element) + 1 in

Allowable bearing pressure in psi: (T = saddle temperature in degrees F)

- for $T \leq 100$ 2,000
- for $100 < T < 500$ $2,000 - 17 (T - 100) / 4$

For details of plate assemblies, refer to Structural Standard Drawing 000.215.4070: Low Friction Slide Assembly - Plans and Sections.

PIER DESIGN**Sizing**

Minimum pier size shall be equal to the saddle (or bearing plate) size plus 4 inches, and adequate to provide lateral anchor bolt cover. Minimum pier thickness shall be 10 inches and should be approximately 10 percent of pier height. Pier dimensions should be in even 2 inch increments.

Reinforcing

Piers should be designed as cantilever beams with two layers of reinforcement. When the required reinforcing approaches ρ_{max} , investigate the pier as a column. Size and reinforcement for each pier should normally be the same. Dowel splices are not required if the vertical pier reinforcing projection is less than 6 feet in height, or the rebar size in feet above the top of the footing. For cases that exceed this limit, use dowels with minimum projections required for tension splices in accordance with ACI 318.

Minimum reinforcing for piers is #5 at 12 inches on each face with #4 ties at 12 inches. Place double ties at top of piers to protect anchor bolts. All ties should encircle the vertical reinforcement. Pier ties are not normally detailed as column ties. If longitudinal reinforcing is not required to resist vertical loads, as is normally the case, through ties are not required.

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Columns (if needed) should be round or rectangular depending on the job criteria or field's preference. Standardize column dimensions as much as possible in order to simplify forming. Column dimensions should be in even 2 inch increments.

Reinforcing

Size and reinforcement for both columns should normally be the same. Use dowels to transfer the column loads to the footings. Minimum dowel projection should be that required for a tension splice in accordance with ACI 318.

FOOTING DESIGN

The following procedure covers footing design for soil supported foundations. For foundations supported on drilled shafts or driven piles, refer to Structural Engineering Guideline: 000.215.1231: Drilled Pile Foundations, and Structural Engineering Guideline: 000.215.1232: Driven Pile Foundations.

Sizing

Plan footing dimensions should be in even 2 inch increments. The footing thickness shall be 12 inches minimum and thickened in 3 inch increments. Size for both footings should normally be the same.

The footing thickness adequate for embedment of pier or column reinforcement should be checked in accordance with ACI 318, Chapter 12. If top tension exists, the footing thickness shall be checked in accordance with ACI 318, Chapter 22 (the curves in Structural Engineering Guideline 000.215.1220, Attachment 03 may be used).

For thin footings with a large concentrated pier moment, the possibility of the moment increasing the punching shear should be considered similar to the way it would be for slabs (refer to ACI 318-99, Section 11.12.2). Engineering judgment should be used in deciding when this might be applicable.

Stability Ratio

The stability ratio (ratio of dead load resisting moment to overturning moment) will be according to job specifications, with a minimum of 1.5 for wind load, and 1.0 for seismic load. When calculating the stability ratio for seismic loads, 90 percent of the vertical dead load should be used in accordance with Structural Engineering Guideline: 000.215.1216: Earthquake Engineering.

Thermal force is an internal force that can be substantially relieved by bending of vessel saddles, rotation of supporting piers or footings, or sliding of saddles on supports. Stability ratio is, therefore, not checked when thermal forces are considered.

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Soil Bearing

Compute soil bearing pressure for footing design and check against the allowable using the following formulae:

Total footing area in compression ($e \leq b/6$):

$$S.B. = \frac{P}{A} \left[1 \pm \left(6 \frac{e}{b} \right) \right] \quad (\text{Equation 3})$$

Total footing area not in compression ($e > b/6$):

$$S.B. = \frac{2 P}{3a (b / 2 - e)} \quad (\text{Equation 4})$$

$$L_b = 3 (b / 2 - e) \quad (\text{Equation 5})$$

where,

- e = Eccentricity of vertical load due to horizontal load
- a = Size of footing perpendicular to direction of horizontal load
- b = Size of footing parallel to direction of horizontal load
- P = Total vertical load (vessel, pier, footing, and soil)
- A = Area of footing
- L_b = footing width in compression

If eccentricity exists in both directions, the above equations don't apply. Numerical solutions can be found in many soil mechanics textbooks. Commercial software is also available for this situation. Attachment 03 is a design aid that is based on accurate numerical solutions and graphically provides the results.

Reinforcing And Stresses

The moment and shear shall be figured on a unit width strip assuming a simple cantilever. The critical section for moment and diagonal tension shear shall be taken at the pier or column face. If shear is excessive, recheck using the critical section for shear specified in ACI 318. The resulting reinforcing steel shall be placed continuously across the entire footing.

If tensile stress in the upper face of footing exceeds the allowable, top steel shall be used provided it is not feasible to increase the footing thickness.

The minimum amount of bottom reinforcing is #5 at 12 inches c/c. If top reinforcing is required, minimum reinforcing is #4 at 12 inches c/c.

REFERENCES

ACI (American Concrete Institute), ACI 318-99, Building Code Requirements for Structural Concrete

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ICBO (International Conference of Building Officials), UBC 97, Uniform Building Code

Structural Engineering

Guideline 000.215.1207: Anchor Bolt Design Criteria

Structural Engineering

Guideline 000.215.1215: Wind Load Calculations

Structural Engineering

Guideline 000.215.1216: Earthquake Engineering

Structural Engineering

Guideline 000.215.1220: Vertical Vessel Foundations

Structural Engineering

Guideline 000.215.1231: Drilled Pile Foundations

Structural Engineering

Guideline 000.215.1232: Driven Pile Foundations

Structural Engineering

Master Specification 000.215.00910: Structural Design Criteria

Structural Engineering

Standard Drawing 000.215.4070: Low Friction Slide Assembly

ATTACHMENTS

Attachment 01: (31Mar05)

Approximate Exchanger and Tube Bundle Weights

Attachment 02: (31Mar05)

Thermal Expansion Data

Attachment 03: (31Mar05)

Soil Pressure For Biaxial Loaded Footings

Attachment 04: (31Mar05)

Sample Design 1: Heat Exchanger Foundation

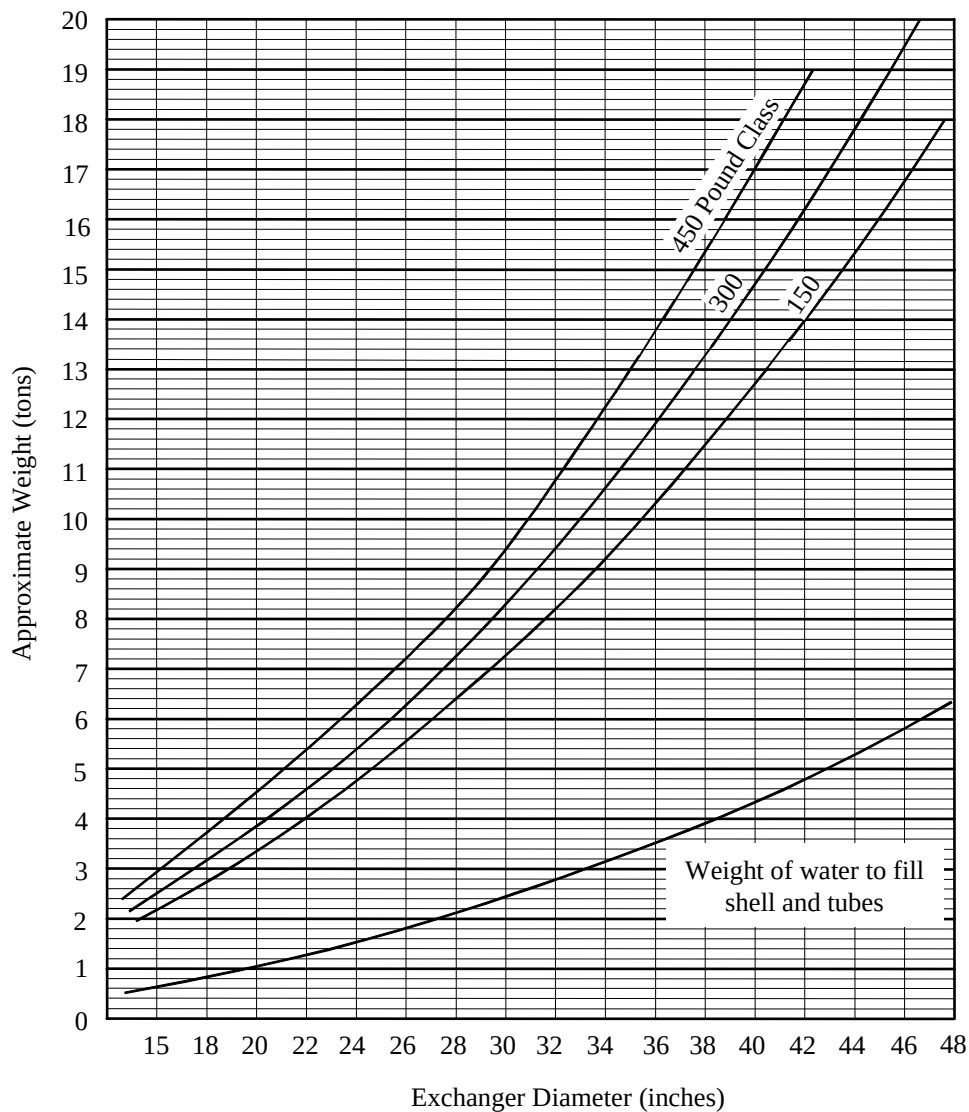
Attachment 05: (31Mar05)

Sample Design 2: Horizontal Vessel Foundation

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Approximate Exchanger And Tube Bundle Weights



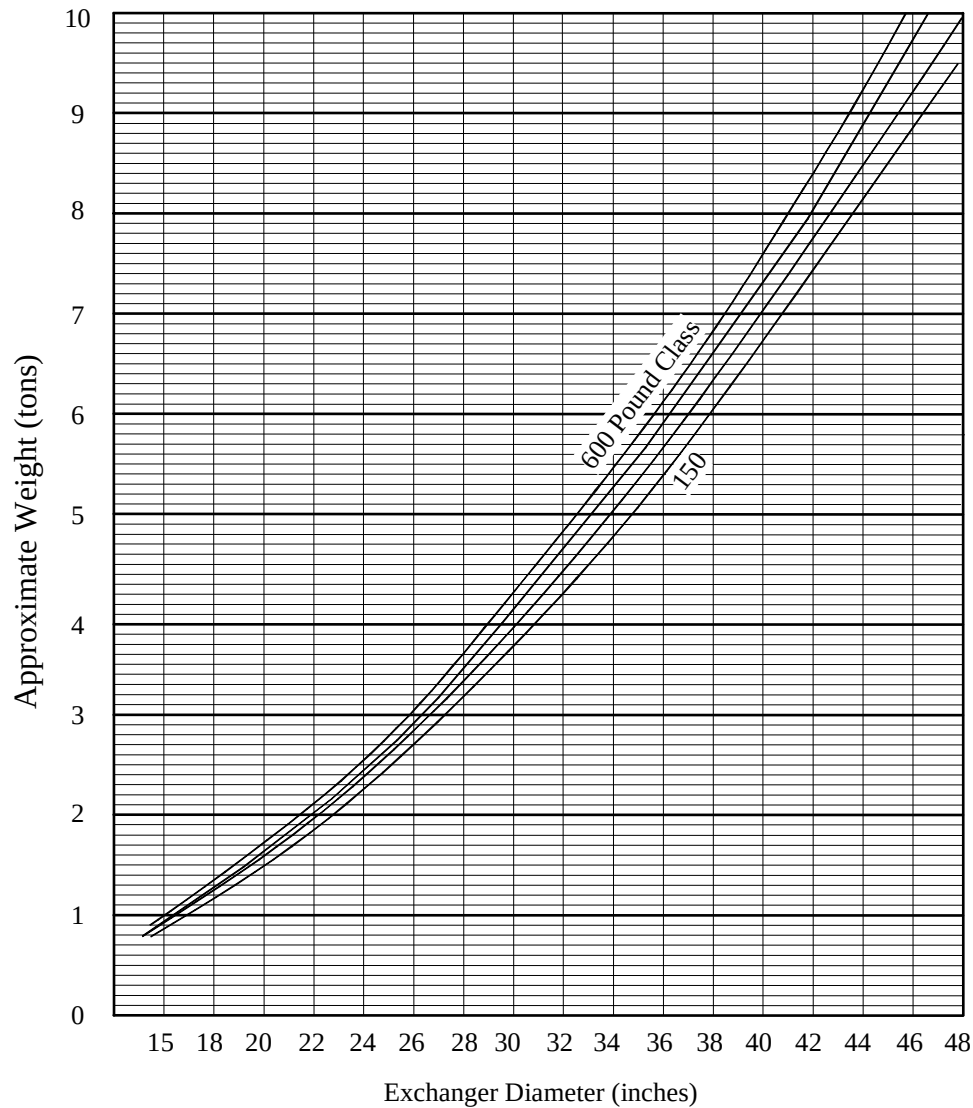
These curves give the approximate weight of standard heat exchangers, all in tons. The curves are for a 192 inch Type ET exchanger with two passes in the tubes. The tubes are 3/4 inch on a 90 degree layout. The tube material is 14 gage steel. For the weights of heat exchangers with other tube lengths, multiply by the following factors:

Length in Inches	240	192	168	144	120	96
Heat Exchanger Factor	1.10	1.00	0.95	0.90	0.85	0.80

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Approximate Exchanger And Tube Bundle Weights



These curves give the approximate weight of standard tube bundles, all in tons. The tubes are 3/4 inch, 14 gage, 192 inches long. Two pass on square pitch. The baffle spacing ranges from 8 inches on the 15 inch exchanger to 16 inches on the 48 inch exchanger. For the weight of bundle with other lengths, multiply by the following factors:

Length in Inches	240	192	168	144	120
Heat Exchanger Factor	1.20	1.00	0.90	0.80	0.70

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Thermal Expansion Data

Total linear expansion between 70 degrees F and indicated temperature (inches/100 feet)

Temp. Degree F	Shell Material								Temp. Degree F
	Carbon Steel Carbon - Moly Low-Chrome (thru 3 Cr Mo)	5 Cr Mo Thru 9 Cr Mo	Austenitic Stainless Steels 18 Cr 8 Ni	12 Cr 17 Cr 27 Cr	12 Cr 20 Ni	Monel 67 Ni 30 Cu	3-1/2 Nickel	Ni-Fe-Cr	
70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70
100	0.23	0.22	0.34	0.20	0.32	0.28	0.23	0.28	100
125	0.42	0.40	0.62	0.36	0.58	0.52	0.42	0.52	125
150	0.61	0.58	0.90	0.53	0.84	0.75	0.61	0.76	150
175	0.80	0.76	1.18	0.69	1.10	0.99	0.81	0.99	175
200	0.99	0.94	1.46	0.86	1.37	1.22	1.01	1.23	200
225	1.21	1.13	1.75	1.03	1.64	1.46	1.21	1.49	225
250	1.40	1.33	2.03	1.21	1.91	1.71	1.42	1.76	250
275	1.61	1.52	2.32	1.38	2.18	1.96	1.63	2.03	275
300	1.82	1.71	2.61	1.56	2.45	2.21	1.84	2.30	300
325	2.04	1.90	2.90	1.74	2.72	2.44	2.05	2.59	325
350	2.26	2.10	3.20	1.93	2.99	2.68	2.26	2.88	350
375	2.48	2.30	3.50	2.11	3.26	2.91	2.47	3.18	375
400	2.70	2.50	3.80	2.30	3.53	3.25	2.69	3.48	400
425	2.93	2.72	4.10	2.50	3.80	3.52	2.91	3.76	425
450	3.16	2.93	4.41	2.69	4.07	3.79	3.13	4.04	450
475	3.39	3.14	4.71	2.89	4.34	4.06	3.35	4.31	475
500	3.62	3.35	5.01	3.08	4.61	4.33	3.58	4.59	500
525	3.86	3.58	5.31	3.28	4.88	4.61	3.81	4.87	525
550	4.11	3.80	5.62	3.49	5.15	4.90	4.04	5.16	550
575	4.35	4.02	5.93	3.69	5.42	5.18	4.27	5.44	575
600	4.60	4.24	6.24	3.90	5.69	5.46	4.50	5.72	600
625	4.86	4.47	6.55	4.10	5.96	5.75	4.74	6.01	625
650	5.11	4.69	6.87	4.31	6.23	6.05	4.98	6.30	650
675	5.37	4.92	7.18	4.52	6.50	6.34	5.22	6.58	675
700	5.63	5.14	7.50	4.73	6.77	6.64	5.46	6.88	700
725	5.90	5.38	7.82	4.94	7.04	6.94	5.70	7.17	725
750	6.16	5.62	8.15	5.16	7.31	7.25	5.94	7.47	750
775	6.43	5.86	8.47	5.38	7.58	7.55	6.18	7.76	775
800	6.70	6.10	8.80	5.60	7.85	7.85	6.43	8.06	800
825	6.97	6.34	9.13	5.82	8.15	8.16	6.68	8.35	825
850	7.25	6.59	9.46	6.05	8.45	8.48	6.93	8.66	850
875	7.53	6.83	9.79	6.27	8.75	8.80	7.18	8.95	875
900	7.81	7.07	10.12	6.49	9.05	9.12	7.43	9.26	900
925	8.08	7.31	10.46	6.71	9.35	9.44	7.68	9.56	925
950	8.35	7.56	10.80	6.94	9.65	9.77	7.93	9.87	950
975	8.62	7.81	11.14	7.17	9.95	10.09	8.17	10.18	975

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Thermal Expansion Data

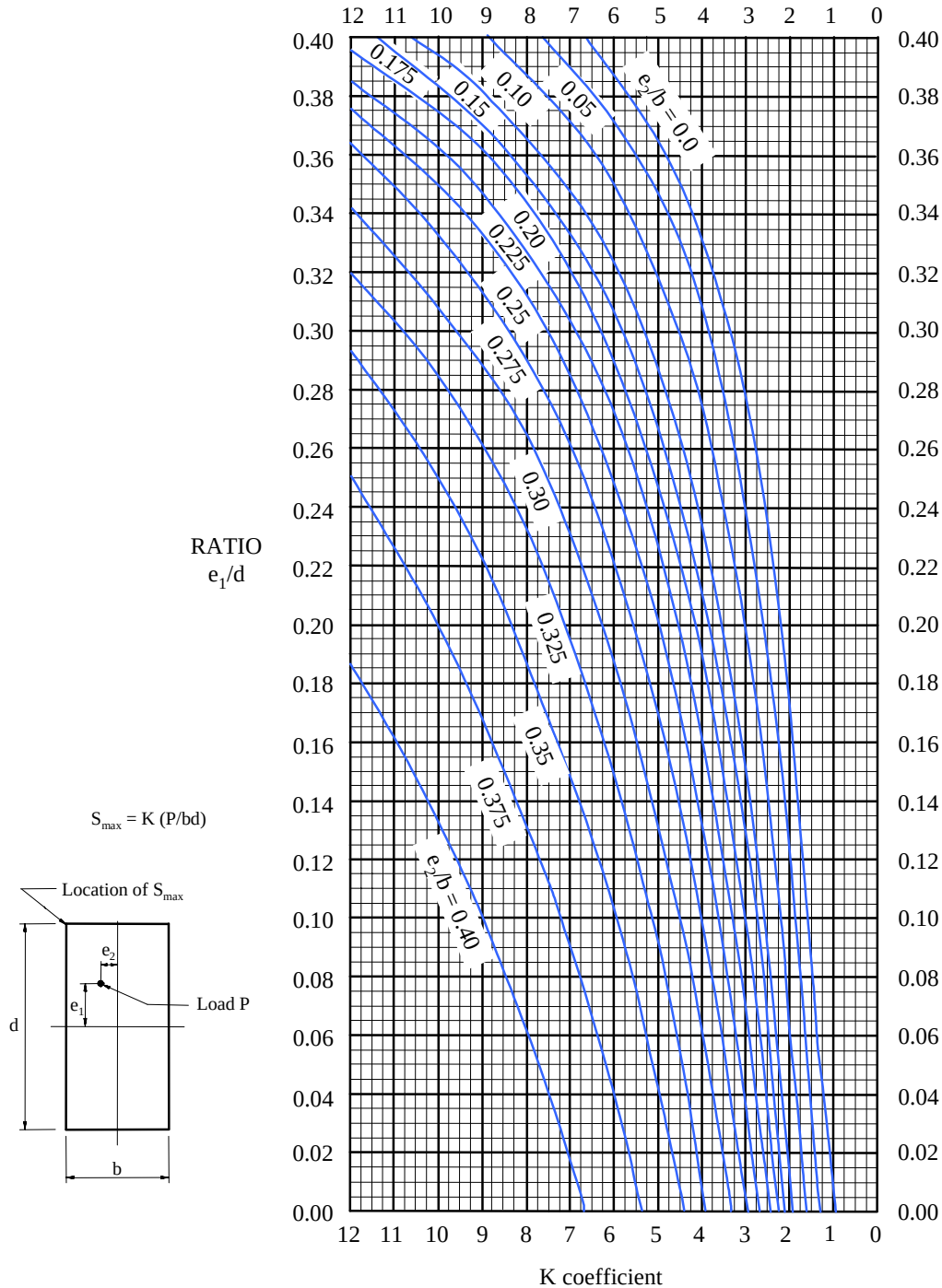
Total linear expansion between 70 degrees F and indicated temperature (inches/100 feet)

Temp. Degree F	Shell Material								Temp. Degree F
	Carbon Steel Carbon - Moly Low-Chrome (thru 3 Cr Mo)	5 Cr Mo Thru 9 Cr Mo	Austenitic Stainless Steels 18 Cr 8 Ni	12 Cr 17 Cr 27 Cr	12 Cr 20 Ni	Monel 67 Ni 30 Cu	3-1/2 Nickel	Ni-Fe-Cr	
1000	8.89	8.06	11.48	7.40	10.25	10.42	8.41	10.49	1000
1025	9.17	8.30	11.82	7.62	10.55	10.75		10.80	1025
1050	9.46	8.55	12.16	7.95	10.85	11.09		11.11	1050
1075	9.75	8.80	12.50	8.18	11.15	11.43		11.42	1075
1100	10.04	9.05	12.84	8.31	11.45	11.77		11.74	1100
1125	10.31	9.28	13.18	8.53	11.78	12.11		12.05	1125
1150	10.57	9.52	13.52	8.76	12.11	12.47		12.38	1150
1175	10.83	9.76	13.86	8.98	12.44	12.81		12.69	1175
1200	11.10	10.00	14.20	9.20	12.77	13.15		13.02	1200
1225	11.38	10.26	14.54	9.42	13.10	13.50		13.36	1225
1250	11.66	10.53	14.88	9.65	13.43	13.86		13.71	1250
1275	11.94	10.79	15.22	9.88	13.76	14.22		14.04	1275
1300	12.22	11.06	15.56	10.11	14.09	14.58		14.39	1300
1325	12.50	11.30	15.90	10.33	14.39	14.94		14.74	1325
1350	12.78	11.55	16.24	10.56	14.69	15.30		15.10	1350
1375	13.06	11.80	16.58	10.78	14.99	15.66		15.44	1375
1400	13.34	12.05	16.92	11.01	15.29	16.02		15.80	1400
1425			17.30					16.16	1425
1450			17.69					16.53	1450
1475			18.08					16.88	1475
1500			18.47					17.25	1500

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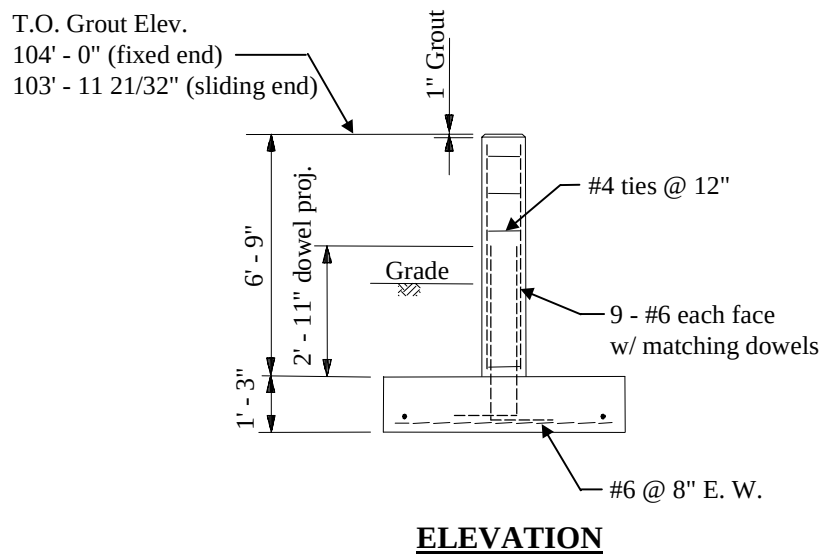
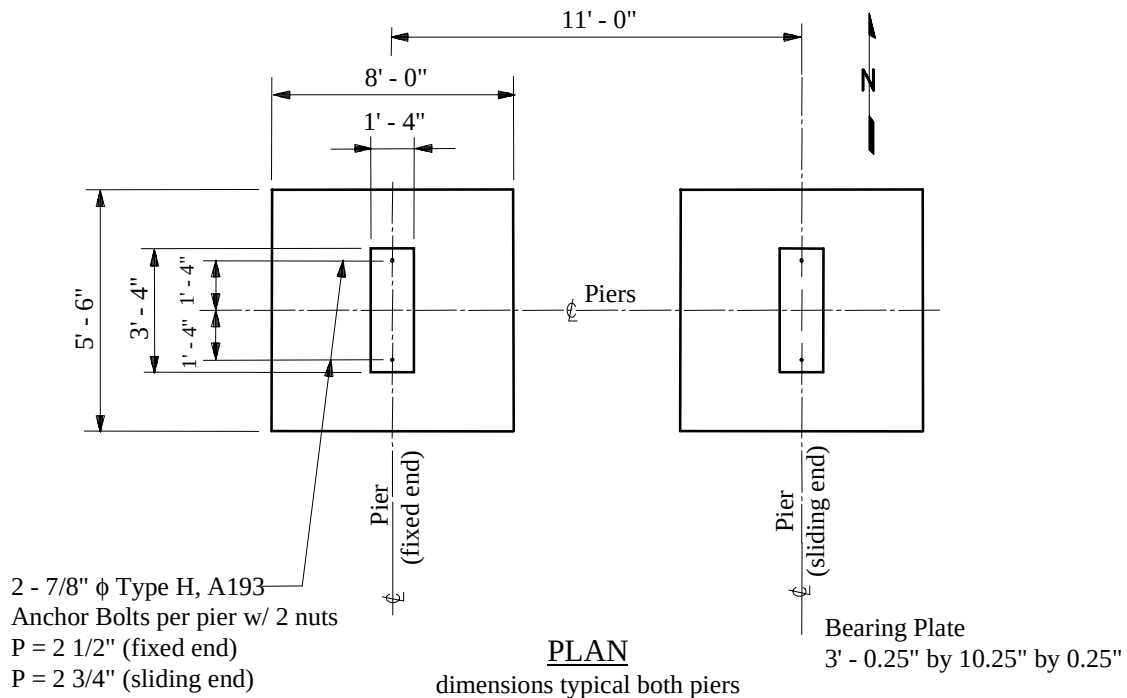
Soil Pressure for Biaxial Loaded Footings



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Sample Design 1: Heat Exchanger Foundation



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Sample Design 1: Heat Exchanger Foundation

DESIGN DATA

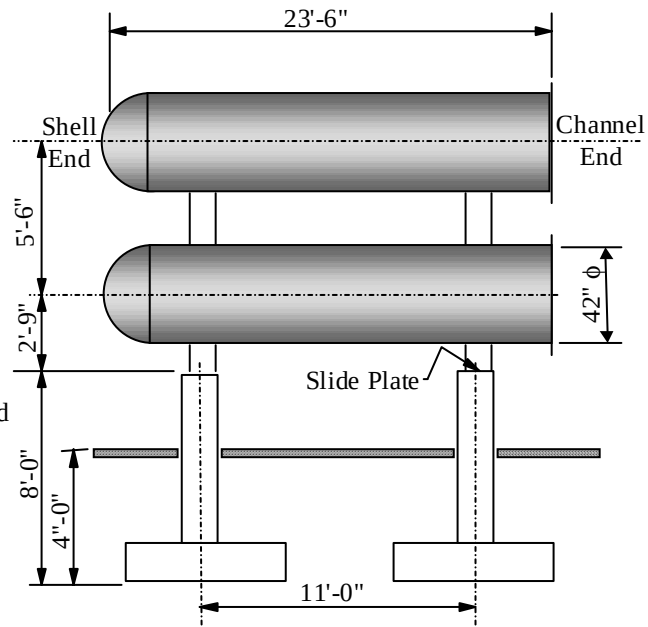
From vessel drawing:

Empty Weight = 32 kips
Operating Weight = 44 kips
Bundle Weight = 19 kips

Basic Diameter = 42 inches, or 3.5 feet
Operating Temperature = 550 degrees F
Vessel Material = carbon steel
Bolts: 2 - 7/8 inch diameter (galvanized)
Bolt Spacing = 2 ft - 8 in c/c
Saddle: 3 ft - 0 in by 9 in
Load Distribution: 60% at channel end, 40% at shell end

From structural criteria specification:

Concrete: $f'_c = 4,000$ psi
Reinforcing: $f_y = 60,000$ psi
Unit weight, $\gamma = 100$ pcf
Net Soil Bearing: $SB_{net} = 5.5$ ksf (at 4 foot depth)



ELEVATION

Wind Load: ASCE 7-95, speed = 100 mph, $I = 1.0$, Exposure C

Seismic Load: UBC 97, zone 2A, soil profile S_D , $I = 1.0$

(in this design, UBC 97 seismic factored load combinations will be used in lieu of ACI 318-99, chapter 9)

DETERMINE LOADS

Empty & Operating Loads

Exchanger weight supplied by outside manufacturers does not include the weight of attached pipes and insulation. Increase exchanger weight by 20% to account for these attached items (refer to Vertical Loads, Empty and Operating Weights).

	60% at channel end	40% at shell end
Empty Weight = $(32 \text{ kips}) + (32 \text{ kips})(1.2) = 70.4 \text{ kips}$	42.2 kips	28.2 kips
Operating Weight = $(44 \text{ kips}) + (44 \text{ kips})(1.2) = 96.8 \text{ kips}$	58.1 kips	38.7 kips

Transverse Moment from Pipe Eccentricity

Eccentricity = $(\text{Basic Diameter}) / 2 + (18 \text{ inches}) = (42 \text{ inches}) / 2 + (18 \text{ inches}) = 39 \text{ inches}$, or 3.25 feet

Empty $M_c = (32 \text{ kips})(0.1)(0.6 \text{ channel end})(3.25 \text{ feet}) = 6.2 \text{ ft-k}$

$M_s = (32 \text{ kips})(0.1)(0.4 \text{ shell end})(3.25 \text{ feet}) = 4.2 \text{ ft-k}$

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Operating $M_c = (44 \text{ kips})(0.1)(0.6 \text{ channel end})(3.25 \text{ feet}) = 8.6 \text{ ft-k}$

$M_s = (44 \text{ kips})(0.1)(0.4 \text{ shell end})(3.25 \text{ feet}) = 5.7 \text{ ft-k}$

Wind Loads

Note: Wind load calculations are not shown for brevity. Refer to Guideline 000.215.1215 for procedures.

Exchanger wind is applied at the center of each exchanger.

Transverse Wind: $H_w = 1.28 \text{ kips}$

Longitudinal Wind: $H_w = 0.25 \text{ kips}$

Transverse or Longitudinal Wind on Pier: $H_w = 0.039 \text{ ksf}$

Seismic Loads

Note: Seismic load calculations are not shown for brevity. Refer to Guideline 000.215.1216 for procedures.

Exchanger seismic loads are applied at the center of each exchanger.

Note that the following are strength level design forces.

	Empty	Operating	Pier
Transverse	5.42 kips	7.45 kips	0.154 W
Longitudinal	8.80 kips	12.10 kips	0.25 W
Vertical	3.87 kips	5.32 kips	0.11 W

For calculations based on allowable stresses, the above forces will be converted to strength-level by dividing by 1.4.

	Empty	Operating	Pier
Transverse	3.87 kips	5.32 kips	0.11 W
Longitudinal	6.29 kips	8.64 kips	0.18 W
Vertical	2.76 kips	3.80 kips	0.08 W

Bundle Pull

$V_{BP} = 1.5 (\text{Bundle Weight}) = 1.5 (19 \text{ k}) = 28.5 \text{ kips}$

Minimum is lessor of 2 kips or exchanger weight. By inspection this will not control

USE $V_{BP} = 28.5 \text{ kips}$ (applied at centerline of top exchanger)

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Thermal Force

1) Compute sliding force: (assume a steel plate is used)

coefficient of friction, $\mu = 0.30$

Operating Load,

$$P_{op} = (2 \text{ exchangers})(44 \text{ kips})(0.6 \text{ channel end}) = 52.8 \text{ kips}$$

$$F_f = \mu (P_{op}) = (0.30)(58.1 \text{ kips at channel end}) = 17.43 \text{ kips} \quad (\text{Equation 1})$$

2) Compute force required to deflect pier:

Assume pier is 40 in wide by 14 in thick by 81 in high

Moment of inertia,

$$I = b(h)^3 / 12 = (40 \text{ in})(14 \text{ in})^3 / 12 = 9,147 \text{ in}^4$$

Modulus of elasticity,

(ACI 318-99, Section 8.5.1)

$$E = 57,000 \sqrt{f'_c} = 57,000 \sqrt{4,000 \text{ psi}} = 3,605 \text{ ksi}$$

Thermal expansion coefficient for carbon steel at 550 degrees F: $\epsilon = 0.0411 \text{ in/ft}$ (Attachment 01)

Thermal growth between piers,

$$\Delta = (\epsilon)(L) = (0.0411 \text{ in/ft})(11 \text{ feet}) = 0.452 \text{ inches}$$

$$F_d = 3 \Delta E I / 2 L^3 = 3 (0.452 \text{ in})(3,605 \text{ ksi})(9,147 \text{ in}^4) / 2 (81 \text{ in})^3 = 42.1 \text{ kips} \quad (\text{Equation 2})$$

Because $F_f < F_d$, and because a lower friction factor won't help the distribution of seismic and bundle pull loads,

USE Steel slide plate.

Size Steel Bearing Plate

$$\text{width} = (\text{saddle width}) + 2(\Delta) + 1/4 \text{ in} = (9 \text{ in}) + 2 (0.452 \text{ in}) + (0.25 \text{ in}) = 10.15 \text{ inches, say } 10.25 \text{ in}$$

$$\text{length} = (\text{saddle length}) + 1/4 \text{ in} = (36 \text{ in}) + (0.25 \text{ in}) = 3 \text{ ft} - 0.25 \text{ in}$$

Check bearing stress: (Operating & Seismic)

$$P_e = (12.10 \text{ kips})(2.75 \text{ ft} + 8.25 \text{ ft}) / (11 \text{ ft between piers}) = 12.10 \text{ kips}$$

$$P_u = 1.1 [1.2 D + 1.0 E] = 1.1 [1.2 (58.1 \text{ kips}) + 1.0 (12.10 \text{ kips})] = 90.00 \text{ kips} \quad (\text{UBC 97, Equation 12-5})$$

$$P_n = \phi 0.85 f'_c A_1 = (0.7)(0.85)(4 \text{ ksi})(10.25 \text{ in})(36.25 \text{ in}) = 884 \text{ kips} > P_u \text{ OK} \quad (\text{ACI 318-99, Section 10.17})$$

USE a bearing plate 3 ft 0 1/4 in by 10 1/4 inches by 1/4 in

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Pier Size

Pier length

$$A) = (c/c \text{ bolts}) + (2)(4 \text{ inch edge distance}) = (2 \text{ ft} - 8 \text{ in}) + 2 (4 \text{ in}) = 3 \text{ ft} - 4 \text{ in}$$

$$B) = (\text{saddle or bearing plate length}) + (4 \text{ in}) = (3 \text{ ft} 0 - 1/4 \text{ in}) + (4 \text{ in}) = 3 \text{ ft} - 4.25 \text{ in} \quad \leftarrow \text{controls}$$

Pier width

$$A) = 10 \text{ inches}$$

$$B) = (\text{saddle or bearing plate width}) + (4 \text{ in}) = (10.25 \text{ in}) + (4 \text{ in}) = 14.25 \text{ inches} \quad \leftarrow \text{controls}$$

USE a pier size of 1 ft - 4 in by 3 ft - 4 in

Anchor Bolt Design

Note: Anchor bolt design is not shown for brevity. Refer to Guideline 000.215.1207 for procedures.

USE 2 – 7/8 in, A193, Type H anchor bolts per pier

Pier Design

at base of pedestal, (assume footing to be 1.25 feet thick)

$$\text{pedestal height} = 8.0 \text{ ft} - 1.25 \text{ ft} = 6.75 \text{ ft}$$

$$\text{pedestal weight} = (0.15 \text{ kcf})(1.33 \text{ ft})(3.33 \text{ ft})(6.75 \text{ ft}) = 4.48 \text{ kips}$$

Operating & Longitudinal Seismic at Fixed End

apply 70% of exchanger seismic at fixed end

$$V_{FX} = 1.1 [(0.7)(12.10 \text{ kips})(2 \text{ exchangers}) + (0.25)(4.48 \text{ kips} / 2)] \quad (\text{UBC 97, Equation 12-5})$$
$$= 19.25 \text{ kips}$$

Empty & Bundle Pull at Fixed End

Factored bundle pull force,

$$V_{BP} = 1.7 (0.75)(28.5 \text{ kips}) = 36.34 \text{ kips}$$

Vertical load due to bundle pull on top exchanger,

$$P_{BP} = (36.34 \text{ kips})(2.75 \text{ ft} + 5.5 \text{ ft}) / (11 \text{ feet between piers}) = 27.26 \text{ kips}$$

Vertical load on sliding pier,

$$P_{SL} = 1.4 (0.75)(42.2 \text{ kips}) - (27.26 \text{ kips}) = 17.05 \text{ kips}$$

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Horizontal load on sliding end,

$$V_{SL} = \mu (P_{SL}) = 0.30 (17.05 \text{ kips}) = 5.12 \text{ kips}$$

Horizontal load on fixed end,

$$V_{FX} = (H_{PB}) - (H_{SL}) = (36.34 \text{ kips}) - (5.12 \text{ kips}) = 31.22 \text{ kips}$$

Empty & Bundle Pull controls,

$$V_u = V_{FX} = 31.22 \text{ kips}$$

$$M_u = (31.22 \text{ kips})(6.75 \text{ ft}) = 210.74 \text{ ft-k}$$

Check diagonal tension shear:

$$d = (16 \text{ in pedestal}) - (2 \text{ in clear}) - (\text{say } 0.75 \text{ in bar}) / 2 = 13.625 \text{ in}$$

$$\begin{aligned} V_n &= 2 \phi \sqrt{f'_c} b d && (\text{ACI 318-99, Equation 11-3}) \\ &= 2 (0.85) \sqrt{4,000 \text{ psi}} (40 \text{ in})(13.625 \text{ in}) / 1,000 = 58.60 \text{ kips} > V_u \quad \text{OK} \end{aligned}$$

select ties from guideline-specified minimum

USE #4 ties @ 12 in

Design for moment:

$$F = b d^2 / 12,000 = (40 \text{ in})(13.625 \text{ in})^2 / 12,000 = 0.62$$

$$K_u = M_u / F = (210.74 \text{ ft-k}) / (0.62) = 340 \rightarrow \rho = 0.0067 \quad (\text{ACI Handbook, Flexure 2.2})$$

$$A_s = \rho b d = (0.0067)(40 \text{ in})(13.625 \text{ in}) = 3.65 \text{ in}^2 \leftarrow \text{controls}$$

The following equation is provided for illustration only, it shouldn't control unless $f'_c > 5,000 \text{ psi}$

$$\begin{aligned} A_{s \min} &= 3 \sqrt{f'_c} b_w d / f_y && (\text{ACI 318-99, Equation 10-3}) \\ &= 3 \sqrt{(4,000 \text{ psi})} (40 \text{ in})(13.625 \text{ in}) / (60,000 \text{ psi}) = 1.72 \text{ in}^2 \end{aligned}$$

$$\begin{aligned} A_{s \min} &= 200 b d / f_y && (\text{ACI 318-99, Section 10.5.1}) \\ &= 200 (40 \text{ in})(13.625 \text{ in}) / (60,000 \text{ psi}) = 1.82 \text{ in}^2 \end{aligned}$$

USE 9 - #6 bars each face ($A_s \text{ provided} = 3.96 \text{ in}^2$)

Determine #6 splice length

$$l_d / d_b = \frac{f_y (\alpha)(\beta)(\lambda)}{25 \sqrt{f'_c}} = \frac{(60,000 \text{ psi})(1.0)(1.0)(0.8)}{25 \sqrt{3,000 \text{ psi}}} = 35.05 \quad (\text{ACI 318-99, Section 12.2.2})$$

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

$$l_d = (35.05) d_b = (35.05)(0.75 \text{ in}) = 26.3 \text{ in}$$

$$\text{Class B splice} = 1.3 (l_d) = 1.3 (26.3 \text{ in}) = 34.19 \rightarrow 35 \text{ in} \quad (\text{ACI 318-99, Section 12.15.1})$$

USE 2 ft - 11 in splice length

Footings Size

Determine minimum footing thickness to develop #6 pedestal reinforcing:

$$l_{hb} = 1200 d_b / \sqrt{f'_c} = 1200 (0.75 \text{ in}) / \sqrt{4,000 \text{ psi}} = 14.2 \text{ in} \quad (\text{ACI 318-99, Section 12.5.2})$$

$$l_{dh} = (14.2 \text{ in})(0.7)(2.73 \text{ in}^2 / 3.08 \text{ in}^2) = 8.81 \text{ in} \quad (\text{ACI 318-99, Section 12.5.3})$$

$$\text{thickness} = (8.81 \text{ in}) + (2 \text{ layers})(0.75 \text{ in rebar}) + (3 \text{ in clear}) = 13.3 \text{ in}$$

USE 15 in minimum

$$SB_{\text{allow}} = (5.5 \text{ ksf net}) + (4.0 \text{ ft deep})(0.10 \text{ kcf soil}) = 5.9 \text{ ksf gross}$$

try a 8 foot by 5.5 foot footing, 1.25 feet thick

Weights:

$$\text{pedestal} = (0.15 \text{ kcf})(3.33 \text{ ft})(1.33 \text{ ft})(8 \text{ ft} - 1.25 \text{ ft}) = 4.48 \text{ kips}$$

$$\text{footing} = (0.15 \text{ kcf})(8 \text{ ft})(5.5 \text{ ft})(1.25 \text{ ft}) = 8.25 \text{ kips}$$

$$\text{soil} = (0.10 \text{ kcf}) [(8 \text{ ft})(5.5 \text{ ft}) - (3.33 \text{ ft})(1.33 \text{ ft})] (4 \text{ ft} - 1.25 \text{ ft}) = 10.88 \text{ kips}$$

$$\text{total} = (4.48 \text{ k}) + (8.25 \text{ k}) + (10.88 \text{ k}) = 23.61 \text{ kips}$$

Check Operating & Longitudinal Seismic & Eccentric (shell end)

$$\text{vertical load due to seismic} = (8.64 \text{ kips})(2.75 \text{ ft} + 8.25 \text{ ft}) / (11.0 \text{ ft}) = 8.64 \text{ kips}$$

$$P = (38.7 \text{ k eqpt}) + (23.61 \text{ k fdn}) \pm (8.64 \text{ kips seismic}) = 53.67 \text{ kips, or } 70.95 \text{ kips}$$

$$M_L = (0.7 \text{ at shell end})(8.64 \text{ k})(2 \text{ exchangers})(8.0 \text{ ft pier}) + (0.18)(4.48 \text{ k pier wt } / 2)(8 \text{ ft}) = 100.0 \text{ ft-kips}$$

$$M_T = 5.7 \text{ ft-kips (ignore)}$$

using maximum axial load,

$$e = M / P = (100.0 \text{ ft-k}) / (70.95 \text{ k}) = 1.41 \text{ ft} > b/6$$

$$SB = 2 P / 3 a [b/2 - e] = 2 (70.95 \text{ k}) / 3 (5.5 \text{ ft}) [(8 \text{ ft}) / 2 - (1.41 \text{ ft})] = 3.32 \text{ ksf} < SB_{\text{allow}} \quad \text{OK}$$

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

using minimal axial load,

$$e = M / P = (100.0 \text{ ft-k}) / (53.67 \text{ k}) = 1.86 \text{ ft} > b/6$$

$$OTM = (100.0 \text{ ft-k}) + (8.64 \text{ k vertical})(8 \text{ ft ftg width}) / 2 = 135 \text{ ft-k}$$

$$RM = 0.9 [(38.7 \text{ eqpt}) + (23.61 \text{ k fdn})] (8 \text{ ft ftg}) / 2 = 224 \text{ ft-k}$$

$$\text{stability ratio} = RM / OTM = (224 \text{ ft-k}) / (135 \text{ ft-k}) = 1.66 > 1.0 \text{ OK}$$

Check Operating & Transverse Seismic & Eccentric (shell end)

$$P = (38.7 \text{ k eqpt}) + (23.61 \text{ k fdn}) = 62.31 \text{ kips}$$

$$M_T = (5.32 \text{ kips}) [(2.75 \text{ ft} + 8.0 \text{ ft}) + (5.5 \text{ ft} + 2.75 \text{ ft} + 8.0 \text{ ft})] * (0.4 \text{ shell end}) + (5.7 \text{ ft-k}) = 63.16 \text{ ft-kips}$$

$$e = M / P = (63.16 \text{ ft-k}) / (62.31 \text{ k}) = 1.01 \text{ feet} > b/6$$

$$SB = 2 P / 3 a [b/2 - e] = 2 (62.31 \text{ k}) / 3 (8 \text{ ft}) [(5.5 \text{ ft}) / 2 - (1.01 \text{ ft})] = 2.98 \text{ ksf} < SB_{\text{allow}} \quad \text{OK} \quad (\text{Equation 4})$$

Check Empty & Bundle Pull & Eccentric (shell end)

vertical load due to bundle pull on top exchanger,

$$P_{BP} = (28.5 \text{ kips})(2.75 \text{ ft} + 5.5 \text{ ft}) / (11 \text{ ft}) = 21.38 \text{ kips}$$

vertical load on sliding end,

$$P_{SL} = (42.2 \text{ kips eqpt}) - (21.38 \text{ kips}) = 20.82 \text{ kips}$$

horizontal load on sliding end,

$$V_{SL} = \mu (P_{SL}) = (0.3)(20.82 \text{ kips}) = 6.25 \text{ kips}$$

horizontal load on fixed end,

$$V_{FX} = (28.5 \text{ kips}) - (6.25 \text{ kips}) = 22.25 \text{ kips}$$

$$P = (28.2 \text{ kips exch}) + (23.61 \text{ kips fdn}) + (21.38 \text{ kips}) = 73.19 \text{ kips}$$

$$M_L = (V_{FX})(\text{fdn height}) = (22.25 \text{ kips})(8.0 \text{ ft}) = 178.0 \text{ ft-kips}$$

$$M_T = 4.2 \text{ ft-kips, (ignore)}$$

$$e = M / P = (178.0 \text{ ft-kips}) / (73.19 \text{ kips}) = 2.43 \text{ feet} > b/6$$

$$SB = 2 P / 3 a [b/2 - e] = 2 (73.19 \text{ k}) / 3 (5.5 \text{ ft}) [(8 \text{ ft}) / 2 - (2.43 \text{ ft})] = 5.65 \text{ ksf} < SB_{\text{allow}} \quad \text{OK}$$

Other cases are OK by inspection

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Determine minimum footing thickness to avoid need for top steel:

depth to bottom of footing = 4.0 feet

length of cantilever = $[(8 \text{ ft}) - (1.33 \text{ ft pier})] / 2 = 3.33 \text{ ft}$

refer to Guideline 000.215.1220, Attachment 03, page 2 → required thickness is less than 12 in, OK

USE 8 foot by 5.5 foot by 1.25 foot footing

Footing Design

Operating & Longitudinal Seismic & Eccentric (shell end):

Due to its minor impact, the transverse eccentric moment will be ignored.

from pier design at top of pier: $V_u = 19.25 \text{ k}$

vertical component of seismic load,

$$P_v = (12.10 \text{ k})(2.75 \text{ ft} + 8.25 \text{ ft}) / (11 \text{ ft}) + 0.11 [38.7 \text{ k eqpt} + 4.48 \text{ k pier} / 2] = 16.60 \text{ kips}$$

$$P_u = 1.1 [1.2 (38.7 \text{ k eqpt} + 23.61 \text{ k fdn}) + 1.0 (16.60 \text{ k seismic})] = 100.51 \text{ kips} \quad (\text{UBC 97, Equation 12-5})$$

$$M_u = (19.25 \text{ k})(8.0 \text{ ft}) = 154.0 \text{ ft-kips}$$

Determine soil resistance:

$$e = M / P = (154.0 \text{ ft-k}) / (100.51 \text{ k}) = 1.53 > b/6$$

using equation 4,

$$SB = 2 b / 3 a (b/2 - e) \\ = 2 (100.51 \text{ k}) / \{3 (5.5 \text{ ft})[(8 \text{ ft}) / 2 - (1.53 \text{ ft})]\} = 4.93 \text{ ksf}$$

using equation 5,

$$\text{bearing length} = 3 (b/2 - e) = 3 [(8 \text{ ft}) / 2 - (1.53 \text{ ft})] = 7.41 \text{ ft}$$

soil bearing at face of pedestal,

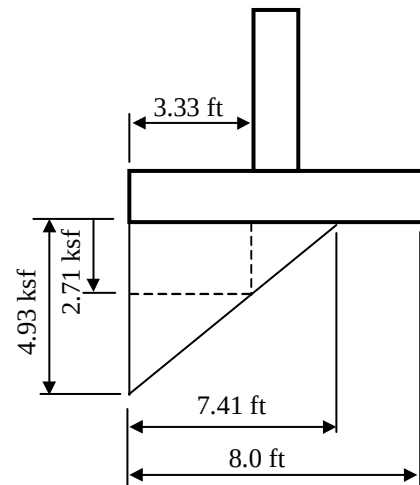
$$SB = (4.93 \text{ ksf})(7.41 \text{ ft} - 3.33 \text{ ft}) / (7.41 \text{ ft}) \\ = 2.71 \text{ ksf (per foot width)}$$

factored soil and concrete weight,

$$w = 1.1 (1.2) [(0.15 \text{ kcf})(1.25 \text{ ft footing}) + (0.1 \text{ kcf})(4.0 \text{ ft soil} - 1.25 \text{ ft})] = 0.61 \text{ ksf}$$

$$V_u = (2.71 \text{ ksf} - 0.61 \text{ ksf})(3.33 \text{ ft}) + (4.93 \text{ ksf} - 2.71 \text{ ksf})(3.33 \text{ ft}) / 2 \\ = 6.99 \text{ kips} + 3.70 \text{ kips} = 10.69 \text{ kips (per foot width)}$$

$$M_u = (6.99 \text{ kips})(3.33 \text{ ft}) / 2 + (3.70 \text{ kips})(3.33 \text{ ft})(2/3) = 19.85 \text{ ft-k (per foot width)}$$



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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Empty & Bundle Pull & Eccentric (shell end):

Due to its minor impact, the transverse eccentric moment will be ignored.

factored bundle pull forces are from pier design.

$$\begin{aligned} P_u &= (\text{exchanger weight}) + (\text{foundation weight}) + (\text{vertical bundle pull force, } P_{BP}) \\ &= 1.4(0.75)[(28.2 \text{ kips exch}) + (23.61 \text{ kips fdn})] + (27.26 \text{ kips}) \\ &= 81.66 \text{ kips} \end{aligned}$$

$$M_u = (\text{horizontal bundle pull, } V_{FX})(\text{fdn height}) = (31.22 \text{ kips})(8.0 \text{ ft}) = 249.76 \text{ ft-kips}$$

Determine soil resistance:

$$e = M_u / P_u = (249.76 \text{ ft-kips}) / (81.66 \text{ kips}) = 3.06 \text{ feet} > b/6$$

$$\begin{aligned} \text{using equation 4,} \\ SB &= 2 P_u / 3 a [b/2 - e] \\ &= 2 (81.66 \text{ k}) / 3 (5.5 \text{ ft}) [(8 \text{ ft}) / 2 - (3.06 \text{ ft})] \\ &= 10.53 \text{ ksf} \end{aligned}$$

$$\begin{aligned} \text{using equation 5,} \\ \text{bearing length} &= 3 (b/2 - e) = 3 [(8 \text{ ft}) / 2 - (3.06 \text{ ft})] \\ &= 2.82 \text{ ft} < \text{distance to face of pedestal} \end{aligned}$$

factored soil and concrete weight,

$$w = 1.4 (0.75) [(0.15 \text{ kcf})(1.25 \text{ ft footing}) + (0.1 \text{ kcf})(4.0 \text{ ft soil} - 1.25 \text{ ft})] = 0.46 \text{ ksf}$$

$$V_u = (10.53 \text{ ksf})(2.82 \text{ ft}) / 2 - (0.46 \text{ ksf})(3.33 \text{ ft}) = 14.85 \text{ kips} - 1.53 \text{ kips} = 13.32 \text{ kips (per foot width)}$$

$$M_u = (14.85 \text{ kips})[(3.33 \text{ ft}) - (2.82 \text{ ft}) / 3] - (1.53 \text{ kips})(3.33 \text{ ft}) / 2 = 32.94 \text{ ft-k (per foot width)}$$

By inspection of results, Empty & Bundle Pull controls

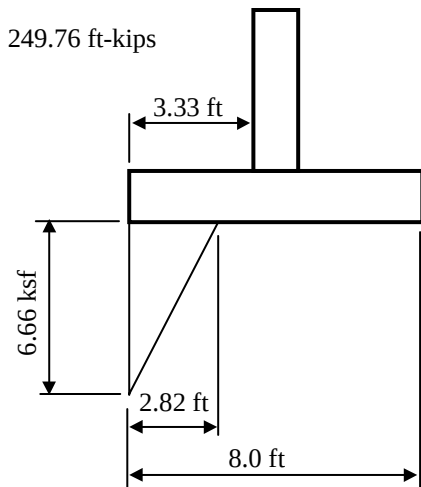
$$V_u = 13.32 \text{ kips}$$

$$M_u = 32.94 \text{ kips}$$

Check diagonal tension shear: (at face of pedestal)

$$d = (15 \text{ in footing}) - (3 \text{ in clear}) - (0.75 \text{ in rebar}) / 2 = 11.63 \text{ in}$$

$$\begin{aligned} V_n &= 2 \phi \sqrt{f'_c} b d && (\text{ACI 318-99, Equation 11-3}) \\ &= 2 (0.85) \sqrt{4,000 \text{ psi}} (12 \text{ in})(11.63 \text{ in}) / 1,000 = 15.01 \text{ kips} > V_u \quad \text{OK} \end{aligned}$$



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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 1: Heat Exchanger Foundation

Design for moment:

$$F = b d^2 / 12,000 = (12 \text{ in})(11.63 \text{ in})^2 / 12,000 = 0.135$$

$$K = M_u / F = (32.94 \text{ ft-k}) / (0.135) = 244 \rightarrow \rho = 0.0047 \quad (\text{ACI Handbook, Flexure 2.2})$$

$$A_s = \rho (b)(d) = (0.0047)(12 \text{ in})(11.63 \text{ in}) = 0.66 \text{ in}^2 \leftarrow \text{controls}$$

$$A_{s \text{ min}} = 200 b d / f_y = 200 (12 \text{ in})(11.69 \text{ in}) / (60,000 \text{ psi}) = 0.47 \text{ in}^2 \quad (\text{ACI 318-99, Section 10.5.1})$$

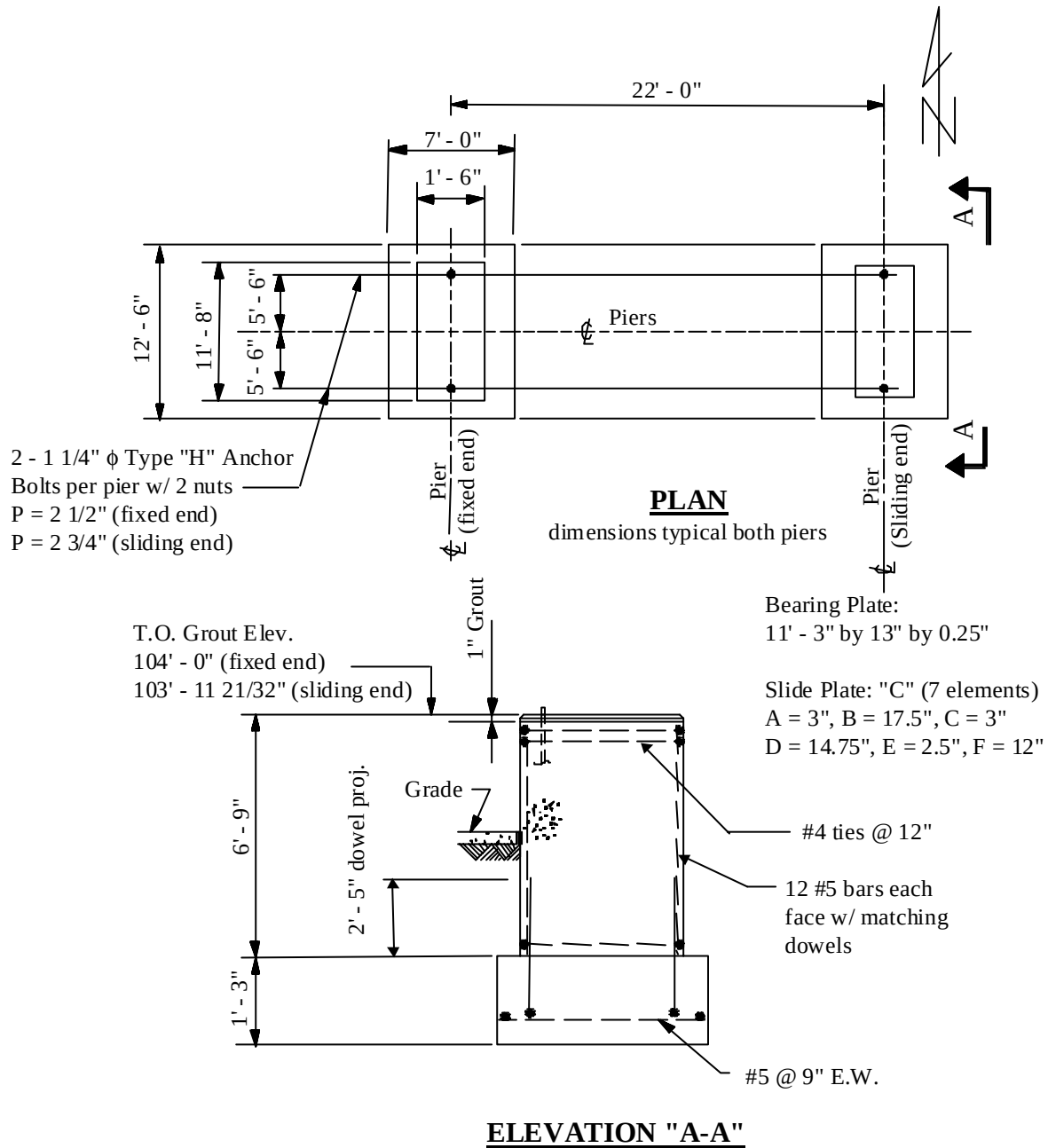
Note: ACI 318-99, Equation 10-3 will not control because $f'_c < 5,000 \text{ psi}$.

USE #6 at 8 in each way ($A_s = 0.66 \text{ in}^2$)

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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation



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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

DESIGN DATA

From vessel drawing:

Empty Weight = 98 kips
Operating Weight = 335 kips
Test Weight = 394 kips

Basic Diameter = 12 feet
Operating Temperature = 500 degrees F
Vessel Material = carbon steel
Bolts: 2 - 1 1/4 in diameter (galvanized)
Bolt Spacing: 11 ft - 0 in
Saddle: 11 ft - 8 in by 10 in

From structural specification criteria:

Concrete: $f'_c = 3,000$ psi
Reinforcing: $f_y = 60,000$ psi
Unit weight: $\gamma = 100$ pcf
Net soil bearing: $SB_{net} = 3.5$ ksf (at 4 foot depth)

Wind Load: ASCE 7-95, speed = 120 mph, $I = 1.15$, Exposure C

Seismic Load: UBC 97, zone 0

Piping load is available and piping eccentric load (as specified in the procedures) is not required.

Use a 1/5 increase in soil allowable pressure for test load combinations.

Wind Loads:

Note: Wind load calculations are not shown for brevity. Refer to Guideline 000.215.1215 for procedures.

Vessel wind is applied at the center of the vessel.

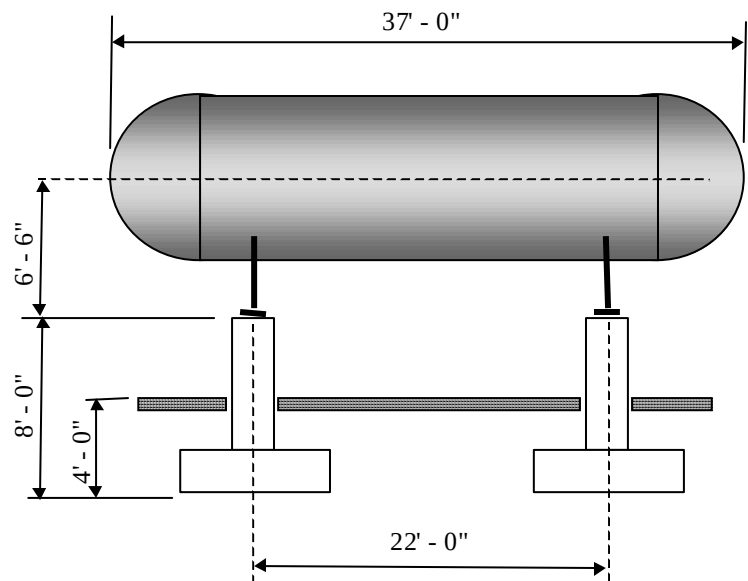
Transverse Wind: $H_w = 13.44$ kips
Longitudinal wind: $H_w = 2.85$ kips

Transverse or Longitudinal wind on pier: $H_w = 0.076$ ksf

Thermal Force:

1) Compute sliding force: (assume a teflon plate used)

by inspection, bearing pressure is greater than 100 psi



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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

Thus, coefficient of friction, $\mu = 0.06$

Operating Load, $P_{op} = (335 \text{ kips}) / (2 \text{ piers}) = 167.5 \text{ kips}$

$F_f = \mu (P_{op}) = (0.06)(167.5 \text{ kips}) = 10.1 \text{ kips}$ (Equation 1)

2) Compute force required to deflect pier:

Assume pier is 12 ft wide by 14 in thick by 78 in high

Moment of inertia, $I = b(h)^3 / 12 = (144 \text{ in})(14 \text{ in})^3 / 12 = 32,928 \text{ in}^4$

Modulus of elasticity, $E = 57,000 \sqrt{f'_c} = 57,000 \sqrt{3,000 \text{ psi}} = 3,122 \text{ ksi}$ (ACI 318-99, Section 8.5.1)

Thermal expansion coefficient for carbon steel at 500 degrees F: $\epsilon = 0.0362 \text{ in/ft}$ (Attachment 01)

Thermal growth between piers, $\Delta = (\epsilon)(L) = (0.0362 \text{ in/ft})(22 \text{ feet}) = 0.796 \text{ inches}$

$F_d = 3 \Delta E I / 2 L^3 = 3 (0.796 \text{ in})(3,122 \text{ ksi})(32,928 \text{ in}^4) / 2 (78 \text{ in})^3 = 258.7 \text{ kips}$ (Equation 2)

Because $F_f < F_d$, and because operating weight at sliding end is greater than 25 kips,

USE Teflon slide plates.

Size Teflon Slide Plate

Bottom element width = (saddle width) + 2 (Δ) + (1/4 in) = (10 in) + 2 (0.796 in) + (0.25 in) $\approx$ 12 inches

Top element width = (bottom element width) + 2(Δ) + (1/2 in) = (12 in) + 2 (0.796 in) + (1 in) $\approx$ 14.75 in

Determine pad length based on allowable bearing stress:

Note: The temperature at the bottom of the saddle can be determined by assuming a 100 degree F per inch temperature gradient for uninsulated steel. For this vessel, with 6 inches between the bottom of the vessel and the bottom of saddle, the potential drop is 600 degrees F. With an operating temperature of 500 degrees F, the bottom of the saddle temperature will be ambient (70 degrees F).

Referring to the guideline's slide plate criteria,
allowable teflon plate bearing stress = 2.0 ksi

Maximum load on sliding end (from Test)

$P_{max} = 394 \text{ kips} / 2 \text{ piers} = 197 \text{ kips}$

Required length of pad based on allowable teflon plate bearing:

length = $P_{max} / (\text{allowable}) = (197 \text{ kips}) / (12 \text{ in})(2 \text{ ksi}) = 8.2 \text{ inches}$

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

USE Assembly "C" with 7 elements

A = 3 in, B = 17.5 in, C = 3 in, D = 14.75 in, E = 2.5 in, F = 12 in

(Standard Drawing 000.215.4070)

Size Steel Bearing Plate

Steel plate dimensions:

$$\text{width} = (\text{bottom teflon element width}) + (1 \text{ in}) = (12 \text{ in}) + (1 \text{ in}) = 13 \text{ inches}$$

$$\text{length} = (11 \text{ ft} - 0 \text{ in c/c bolts}) + (3 \text{ in}) = 11 \text{ ft} - 3 \text{ in}$$

Check bearing stress: (Test)

$$P_u = 1.4 D = 1.4 (394 \text{ kips} / 2 \text{ piers}) = 275.8 \text{ kips}$$

$$P_n = \phi 0.85 f'_c A_1 = (0.7)(0.85)(3 \text{ ksi})(13 \text{ in})(135 \text{ in}) = 3,132 \text{ kips, OK} \quad (\text{ACI 318-99, Section 10.17})$$

USE a bearing plate 11 ft 3 in by 13 inches by 0.25 in

Pier Size

Pier length

$$A) = (\text{c/c bolts}) + (2)(4 \text{ inch edge distance}) = (11 \text{ ft} - 0 \text{ in}) + 2 (4 \text{ in}) = 11 \text{ ft} - 8 \text{ in} \quad \leftarrow \text{controls}$$

$$B) = (\text{saddle or bearing plate length}) + (4 \text{ in}) = (11 \text{ ft} - 3 \text{ in}) + (4 \text{ in}) = 11 \text{ ft} - 7 \text{ in}$$

Pier width

$$A) = 10 \text{ inches}$$

$$B) = (\text{saddle or bearing plate width}) + (4 \text{ in}) = (13 \text{ in}) + (4 \text{ in}) = 17 \text{ inches} \quad \leftarrow \text{controls}$$

USE a pier size of 1 ft - 6 in by 11 ft - 8 in

Anchor Bolt Design

Note: Anchor bolt design is not shown for brevity. Refer to Guideline 000.215.1207 for procedures.

USE 2 - 1 1/4 in, A36, Type H anchor bolts per pier

Pier Design

By inspection of loads, maximum moment in pier will be from Operating & Thermal

Determine factored forces:

at base of pedestal, (assume footing to be 1.25 feet thick)

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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

$$\text{pedestal height} = 8.0 \text{ ft} - 1.25 \text{ ft} = 6.75 \text{ ft}$$

$$V_u = 1.4 (10.1 \text{ kips}) = 14.14 \text{ kips} \quad (\text{ACI 318-99, Equation 9-1})$$

$$M_u = (14.14 \text{ kips})(6.75 \text{ ft}) = 95.45 \text{ ft-k}$$

Check diagonal tension shear:

$$d = (18 \text{ in pedestal}) - (2 \text{ in clear}) - (\text{say } 0.75 \text{ in bar}) / 2 = 15.625 \text{ in}$$

$$\begin{aligned} V_n &= 2 \phi \sqrt{f'_c} b d & (\text{ACI 318-99, Equation 11-3}) \\ &= 2 (0.85) \sqrt{3,000 \text{ psi}} (140 \text{ in})(15.625 \text{ in}) / 1,000 = 203.68 \text{ kips} > V_u \quad \text{OK} \end{aligned}$$

select ties from guideline-specified minimum

USE #4 ties @ 12 in

Design for moment:

$$F = b d^2 / 12,000 = (140 \text{ in})(15.625 \text{ in})^2 / 12,000 = 2.85$$

$$K_u = M_u / F = (95.45 \text{ ft-k}) / (2.85) = 33 \rightarrow \rho \approx 0.0010 \quad (\text{ACI Handbook, Flexure 2.2})$$

$$A_s = \rho b d = (0.0010)(140 \text{ in})(15.625 \text{ in}) = 2.19 \text{ in}^2$$

Note: ACI 318-99, Equation 10-3 will not control because $f'_c < 5,000 \text{ psi}$.

$$\begin{aligned} A_{s \text{ min}} &= 200 b d / f_y & (\text{ACI 318-99, Section 10.5.1}) \\ &= 200 (140 \text{ in})(15.625 \text{ in}) / (60,000 \text{ psi}) = 7.29 \text{ in}^2 \end{aligned}$$

$$A_{s \text{ min}} = (4/3)(A_s) = (4/3)(2.19 \text{ in}^2) = 2.92 \text{ in}^2 \leftarrow \text{controls}$$

by inspection, the guideline-defined minimum of #5 at 12 in will be used.

USE 12- #5 bars each face (A_s provided = 3.41 in²)

Determine #5 splice length

$$l_d / d_b = \frac{f_y (\alpha)(\beta)(\lambda)}{25 \sqrt{f'_c}} = \frac{(60,000 \text{ psi})(1.0)(1.0)(0.8)}{25 \sqrt{3,000 \text{ psi}}} = 35.05 \quad (\text{ACI 318-99, Section 12.2.2})$$

$$l_d = (35.05) d_b = (35.05)(0.625 \text{ in}) = 21.9 \text{ in}$$

$$\text{Class B splice} = 1.3 (l_d) = 1.3 (21.9 \text{ in}) = 28.47 \rightarrow 29 \text{ in} \quad (\text{ACI 318-99, Section 12.15.1})$$

FLUOR

HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

USE 2 ft - 5 in splice length

Footing Size

Determine minimum footing thickness to develop #5 pedestal reinforcing:

$$l_{hb} = 1200 d_b / \sqrt{f'_c} = 1200 (0.625 \text{ in}) / \sqrt{3,000 \text{ psi}} = 13.7 \text{ in} \quad (\text{ACI 318-99, Section 12.5.2})$$

$$l_{dh} = (13.7 \text{ in})(0.7)(2.92 \text{ in}^2 / 3.41 \text{ in}^2) = 8.21 \text{ in} \quad (\text{ACI 318-99, Section 12.5.3})$$

$$\text{thickness} = (8.21 \text{ in}) + (2 \text{ layers})(0.625 \text{ in rebar}) + (3 \text{ in clear}) = 12.46 \text{ in}$$

USE 15 in minimum

$$SB_{\text{allow}} = (3.5 \text{ ksf net}) + (4.0 \text{ ft deep})(0.10 \text{ kcf soil}) = 3.9 \text{ ksf gross}$$

try a 12.5 foot by 7 foot footing, 1.25 feet thick

Weights:

$$\text{pedestal} = (0.15 \text{ kcf})(11.67 \text{ ft})(1.5 \text{ ft})(8 \text{ ft} - 1.25 \text{ ft}) = 17.72 \text{ kips}$$

$$\text{footing} = (0.15 \text{ kcf})(12.5 \text{ ft})(7.0 \text{ ft})(1.25 \text{ ft}) = 16.41 \text{ kips}$$

$$\text{soil} = (0.10 \text{ kcf}) [(12.5 \text{ ft})(7.0 \text{ ft}) - (11.67 \text{ ft})(1.5 \text{ ft})] (4 \text{ ft} - 1.25 \text{ ft}) = 19.25 \text{ kips}$$

$$\text{total} = (17.72 \text{ k}) + (16.41 \text{ k}) + (19.25 \text{ k}) = 53.38 \text{ kips}$$

Check Operating & Thermal

$$P = (335 \text{ k} / 2 \text{ piers}) + (53.38 \text{ k fdn}) = 220.88 \text{ kips}$$

$$M_L = (10.1 \text{ k})(8.0 \text{ ft}) = 80.8 \text{ ft-kips}$$

$$e = M / P = (80.8 \text{ ft-k}) / (220.88 \text{ k}) = 0.37 < b/6$$

$$SB = P / A [1 + 6e / b] = [(220.88 \text{ k}) / (12.5 \text{ ft})(7 \text{ ft})] [1 + 6 (0.37) / (7 \text{ ft})] = 3.3 \text{ ksf} < SB_{\text{allow}} \quad \text{OK} \quad (\text{Equation 3})$$

stability ratio is OK by inspection.

Check Operating & Transverse Wind

$$P = (335 \text{ k} / 2 \text{ piers}) + (53.38 \text{ k fdn}) = 220.88 \text{ kips}$$

$$M_T = (13.44 \text{ kips})(8 \text{ ft} + 6.5 \text{ ft}) = 194.88 \text{ ft-kips}$$

$$e = M / P = (194.88 \text{ ft-k}) / (220.88 \text{ k}) = 0.88 \text{ feet} < b/6$$

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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

$$SB = (P / A) [1 + 6e / b]$$

$$= [(220.88 \text{ k}) / (12.5 \text{ ft})(7 \text{ ft})] [1 + 6 (0.88) / (12.5 \text{ ft})] = 3.6 \text{ ksf} < SB_{\text{allow}} \quad \text{OK} \quad (\text{Equation 3})$$

Other cases are OK by inspection

Determine minimum footing thickness to avoid need for top steel:

Check using the guideline chart without regard to actual amount of uplift. If this doesn't work, use a less conservative method.

depth to bottom of footing = 4.0 feet

length of cantilever = $[(7 \text{ ft}) - (1.5 \text{ ft pier})] / 2 = 2.75 \text{ ft}$

refer to Guideline 000.215.1220, Attachment 03, page 2 → required thickness is less than 12 in, OK

USE 12.5 foot by 7.0 foot by 1.25 foot footing

Footing Design

By inspection of loads, maximum moment in footing will be from Operating and Thermal

Determine factored forces: (ACI 318-99, Equation 9-1)

from pier design at bottom of pier: $V_u = 14.14 \text{ k}$, $M_u = 95.45 \text{ k}$

$P_u = 1.4 [(53.38 \text{ k fdn weight}) + (335 \text{ k eqpt weight}) / (2 \text{ piers})] = 309.23 \text{ kips}$

$M_u = (95.45 \text{ ft-k}) + (14.14 \text{ k})(1.25 \text{ ft footing}) = 113.13 \text{ ft-kips}$

Determine soil resistance:

$$e = M / P = (113.13 \text{ ft-k}) / (309.23 \text{ k}) = 0.37 < b/6$$

using equation 3,

$$SB = (P / A) [1 \pm 6e/b]$$

$$= [(309.23 \text{ k}) / (12.5 \text{ ft})(7 \text{ ft})] [1 \pm 6(0.37) / (7 \text{ ft})]$$

$$= 4.65 \text{ ksf}, 2.41 \text{ ksf}$$

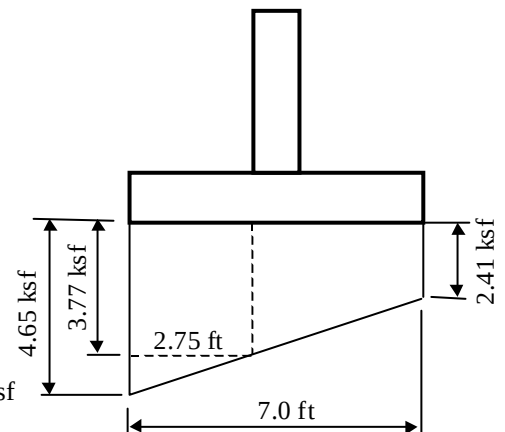
at face of pedestal,

$$SB = (4.65 \text{ ksf}) - (2.75 \text{ ft})(4.65 \text{ ksf} - 2.41 \text{ ksf}) / (7 \text{ ft}) = 3.77 \text{ ksf}$$

$$V_u = (3.77 \text{ ksf})(2.75 \text{ ft}) + (4.65 \text{ ksf} - 3.77 \text{ ksf})(2.75 \text{ ft}) / 2$$

$$= 10.37 \text{ k} + 1.21 \text{ k} = 11.58 \text{ kips (per foot width)}$$

$$M_u = (10.37 \text{ k})(2.75 \text{ ft}) / 2 + 2 (1.21 \text{ k}) / 3 = 15.07 \text{ ft-k (per foot width)}$$



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HEAT EXCHANGER AND HORIZONTAL VESSEL FOUNDATIONS

Sample Design 2: Horizontal Vessel Foundation

Check diagonal tension shear: (at face of pedestal)

$$d = (15 \text{ in footing}) - (3 \text{ in clear}) - (0.625 \text{ in rebar}) / 2 = 11.69 \text{ in}$$

$$\begin{aligned} V_n &= 2 \phi \sqrt{f'_c} b d && (\text{ACI 318-99, Equation 11-3}) \\ &= 2 (0.85) \sqrt{3,000 \text{ psi}} (12 \text{ in})(11.69 \text{ in}) / 1,000 = 13.06 \text{ kips} > V_u \quad \text{OK} \end{aligned}$$

Design for moment:

$$F = b d^2 / 12,000 = (12 \text{ in})(11.69 \text{ in})^2 / 12,000 = 0.137$$

$$K = M_u / F = (15.07 \text{ ft-k}) / (0.137) = 110 \rightarrow \rho = 0.0021 \quad (\text{ACI Handbook, Flexure 2.2})$$

$$A_s = \rho b d = (0.0021)(12 \text{ in})(11.69 \text{ in}) = 0.29 \text{ in}^2$$

Note ACI 318-99, Equation 10-3 will not control because $f'_c < 5,000 \text{ psi}$.

$$A_{s \min} = 200 b d / f_y = 200 (12 \text{ in})(11.69 \text{ in}) / (60,000 \text{ psi}) = 0.47 \text{ in}^2 \quad (\text{ACI 318-99, Section 10.5.1})$$

$$A_{s \min} = (4/3)(A_s) = (4/3)(0.29 \text{ in}^2) = 0.39 \text{ in}^2 \leftarrow \text{controls}$$

USE #5 at 9 in each way ($A_s = 0.41 \text{ in}^2$)